

Randomized and low-rank approximation

# HSS approximation from a number of matvecs independent of depth

**Difficulty:** challenging **Importance:** interesting to the community**Status:** open; checked 2026-09-08**Area:** randomized matrix compression

## Context and notation

Use exact real arithmetic, with comparisons, standard Gaussian sampling, and exact SVDs available as primitives; charge an SVD of an  $a \times b$  matrix  $O(ab \min(a, b))$  operations. This states the idealized arithmetic model used here, rather than a finite precision or bit complexity claim. A matrix–vector query returns either  $Av$  or  $A^\top v$  for one chosen real vector  $v$ ; both types count toward the total. Randomized guarantees are for every fixed input, with probability or expectation over the algorithm’s randomness.

For  $N = 2^{L+1}k$ ,  $k, L \geq 1$ , form the perfect binary tree that successively halves the ordered index set  $[N] = \{1, \dots, N\}$  until every leaf contains  $2k$  indices. Define  $\mathcal{S}_{L,k}$  to contain precisely the real  $N \times N$  matrices  $B$  such that, for every nonroot node  $I$  of this tree,

$$\text{rank } B[I, [N] \setminus I] \leq k, \quad \text{rank } B[[N] \setminus I, I] \leq k.$$

This is the HSS class with rank at most  $k$ , including its constraints across levels. There is no restriction inside a leaf diagonal block. Set

$$E_{L,k}(A) = \min_{B \in \mathcal{S}_{L,k}} \|A - B\|_F^2.$$

The minimum exists. The rank definition follows Stefano Masei, Leonardo Robol, and Daniel Kressner, *hm-toolbox: MATLAB Software for HODLR and HSS Matrices*, SIAM Journal on Scientific Computing **42** (2020), C43–C68, §2.2, Definition 3. Equivalence with the telescoping representation and existence of a minimizer are recorded in Amsel et al., cited below, Definition 3 and Appendix D.

## Problem statement

Do absolute constants  $C_q, C_e, c > 0$  and a uniform randomized algorithm exist that, for every  $k, L \geq 1$  and every  $A \in \mathbb{R}^{N \times N}$ ,  $N = 2^{L+1}k$ , accesses  $A$  solely through at most  $C_q k$  matrix–vector queries and returns  $B \in \mathcal{S}_{L,k}$  satisfying

$$\mathbb{E} \|A - B\|_F^2 \leq C_e L E_{L,k}(A),$$

with at most  $O(N^c)$  additional arithmetic operations? Queries may be adaptive. The budget counts both transpose and forward products and is a bound on every execution.

## References

Amsel et al., *Quasi-optimal Hierarchically Semi-separable Matrix Approximation*, §1.2, Theorem 13, §§4.4–5. James Levitt and Per-Gunnar Martinsson, *Linear-Complexity Black-Box Randomized Compression of Rank-Structured Matrices*, SIAM Journal on Scientific Computing **46** (2024), A1747–A1763, Algorithm 4.1 and Remark 4.3.

### Status evidence

The proved guarantee uses  $O(kL)$  queries. Reusing sketches reduces the query count, but its approximation guarantee remains unproved. Searches for “HSS  $O(k)$  approximation 2026 matvec” and “HSS approximation logarithmic 2026” found no resolution. Levitt–Martinsson’s latest arXiv version is v3, submitted June 21, 2024. Christopher Musco’s [February 2026 ICERM slides](#), numbered slide 14, still list  $O(k \log N)$  queries for HSS approximation.

[RE-01](#) permits full access and asks for a constant error factor. RE-02 retains the known dependence of the error factor on depth and asks for fewer queries. Neither target directly implies the other.